Spatio-temporal Assessment of Nighttime Light Emissions and Urban Vulnerability: A Case Study of Dhaka for the year 2015, 2020 and 2025

Introduction

Dhaka, the capital of Bangladesh, has experienced rapid urbanization in recent decades. Remote sensing indices such as NDBI (Normalized Difference Built-up Index) and NTL (Nighttime Lights) provide effective proxies to measure physical expansion and socio-economic activity at night. But these extreme amount of light at nights is considered to be pollution. Nighttime light pollution harms people by disrupting sleep, suppressing melatonin, and increasing risks of cancer, metabolic disorders, and mental health problems.

Goal and Objectives

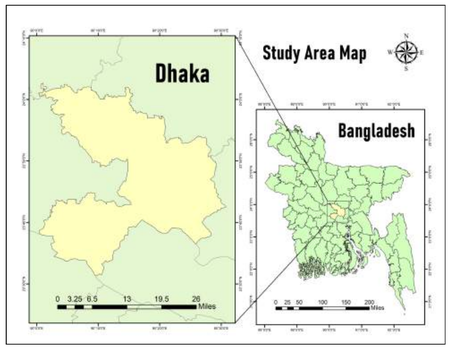
Goal:
To develop a Vulnerability Map for Nighttime lights while analyzing Nighttime Light Emission Trend and Assessing its Correlation with Built up Intensity in Dhaka for the year 2015, 2020 and 2025.

Objectives:

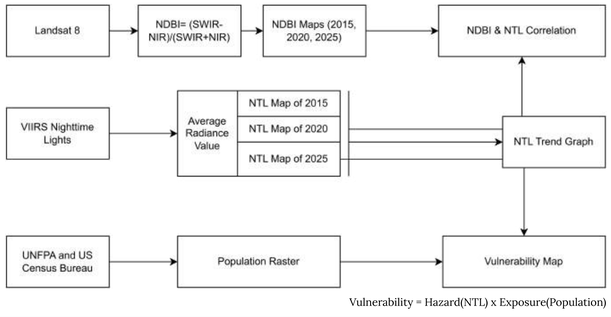
- Derive NDBI maps for 2015, 2020, 2025
- Generate NTL maps and showing trend for these 3 years.
- Compare NDBI vs NTL using scatter plots and interpret correlation between them.
- Develop a vulnerability map combining hazard (NTL) and exposure (population) and finding out the most vulnerable zones.

Study Area

Dhaka, the capital of Bangladesh, faces intense urban expansion and rising population density, making it a key area for analyzing built-up growth, nighttime activity, and vulnerability.

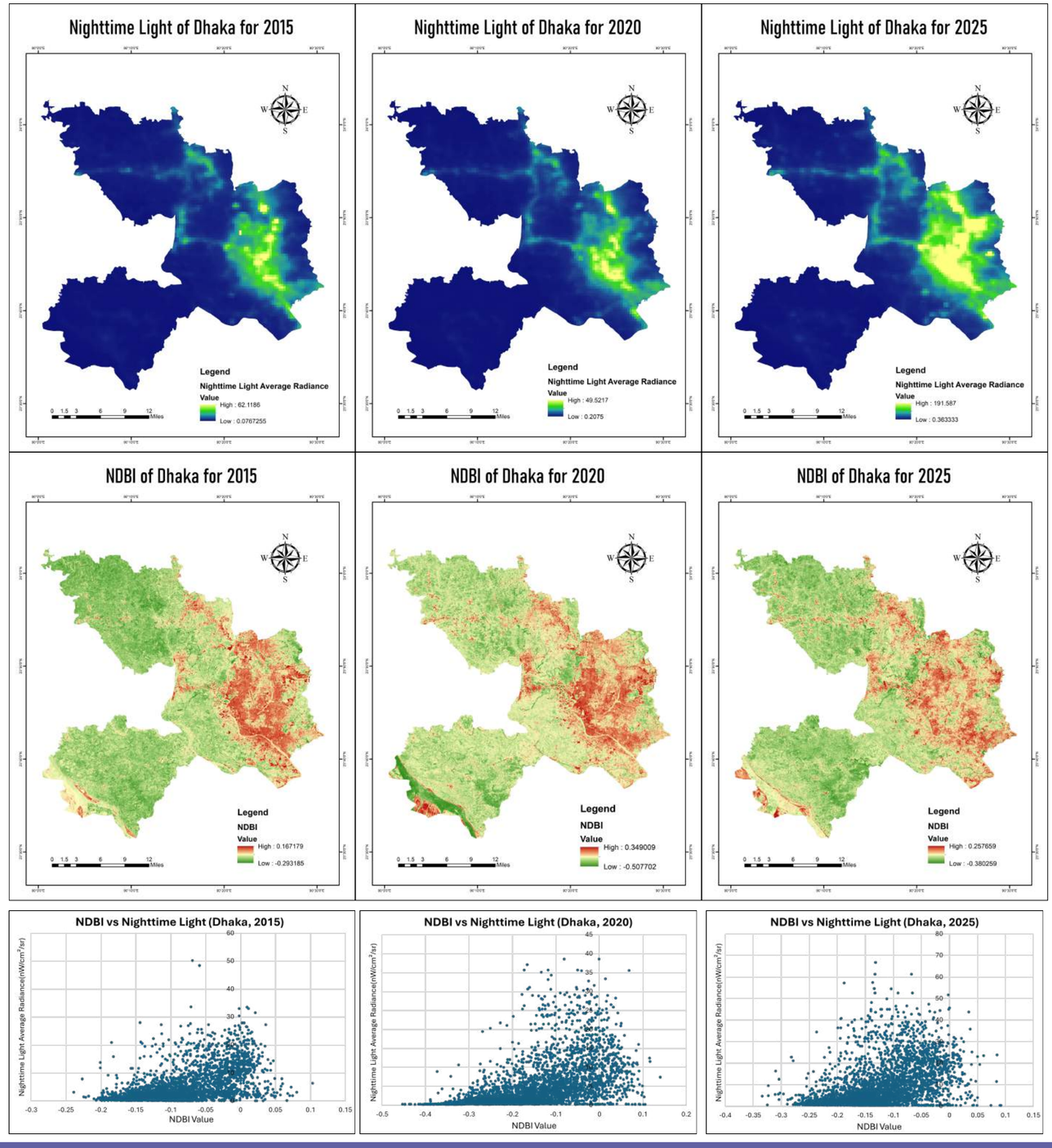


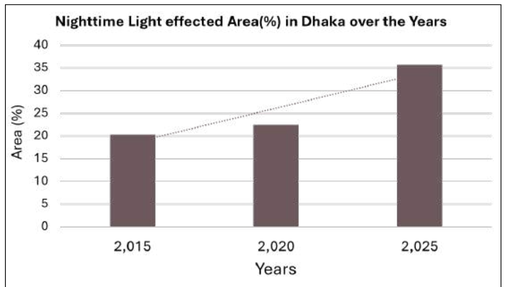
Methodology

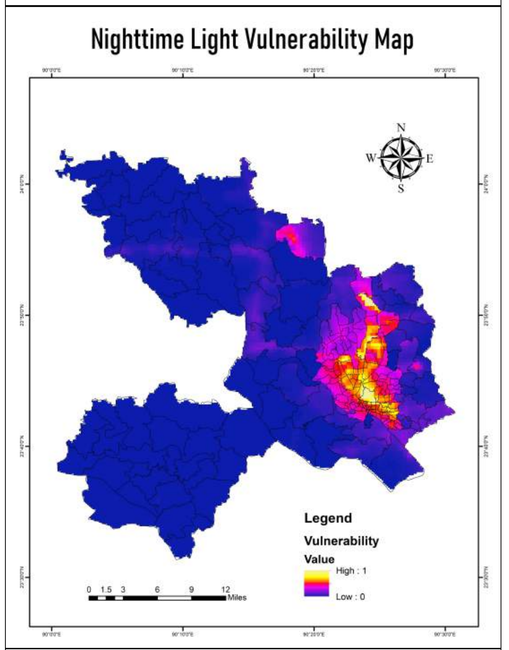


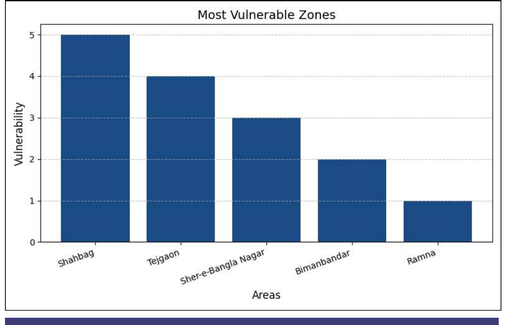
Vulnerability = Hazard(NTL) x Exposure(Population)

Analysis









Findings

- Steady NTL Expansion from 2015 to 2025
- Strong Correlation between built-up intensity and nighttime light emission.
- Shahbag, Tejgaon, Sher-e-Bangla Nagar, Ramna, Bimanbandar these are the most Vulnerable aeras for Nighttime light emissions in Dhaka.

Submitted by : ID- 2105051
Course No: URP 364
Course Name : Remote Sensing and Photogrammetry Lab

Submitted to:
Debasish Roy Raja (Associate Professor, DURP, CUET)
Md. Kamrul Islam (Assistant Professor, DURP, CUET)

DEPARTMENT OF URBAN AND REGIONAL PLANNING
CHITTAGONG UNIVERSITY OF ENGINEERING & TECHNOLOGY
CHATTOGRAM – 4349, BANGLADESH